LEARNING AND WORKING OUR WAY TO SUCCESS

Practice is making perfect as Home Performance with Energy Star branches out into new territory.

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As part of the comprehensive evaluation of the home, contractors participating in Home Performance with Energy Star provide a variety of safety checks, including verifying proper and effective venting of combustion equipment and even testing for gas leaks.

BY MIKE ROGERS

Four years after the launch of Home Performance with Energy Star—a program aimed at bringing home performance to existing homes—I am excited first and foremost by its success in the marketplace. In New York, where the program first took root, over 5,800 homeowners have invested more than $40 million of their own money in home performance improvements—improvements that were completed by qualified contractors using advanced diagnostics to make these homes more energy efficient, safer, and more comfortable (see "A New York Energy $mart Program," p. 24). In Wisconsin, the number of completed home performance jobs has increased tenfold in the three years of the program's existence there (see "Wisconsin's Home Performance with Energy Star," p. 25). While homeowners are suddenly finding themselves comfortable in their own homes, contractors are also benefiting from the program. Contractors who incorporate this comprehensive, whole-house approach into their businesses are finding they can charge a fair price, make a decent living, and feel good about the difference they're making to their customers.

Because home performance contracting makes sense for contractors and homeowners alike, this whole-house approach to home improvement works even in markets that have no program support. Contractors like Joe Kuonen and Royce Lewis in Arkansas, and Mike Woodson and Jerold Sit in Mississippi, are demonstrating this on a daily basis (see "The Right Way Is Right," *HE* Nov/Dec '04, p. 40). Unfortunately, for a variety of reasons, contractors are only figuring out the enormous opportunities at their fingertips one at a time. There are tens of millions of homes that could use some performance and efficiency improvements. Yet right now most homeowners don't have ready access to contractors who understand the science behind how the whole house works or how to apply that knowledge. That's where Home Performance with Energy Star comes in. As the program spreads across the country, we are learning how to connect the dots more quickly between trained contractors and customers.

As the two programs with the longest history in Home Performance with Energy Star, New York and Wisconsin provide a long-term view of how this program can most effectively make an impact. Hats are off to the two programs for jumping in early and sharing their experiences. They also illustrate that not all Home Performance programs have to be identical to work well; these two programs operate using different contractor business models. And we are continuing to learn from the newer programs arising in California; Kansas City and St. Louis, Missouri; Austin, Texas; Atlanta, Georgia; and elsewhere. Each is contributing to our understanding of how to help make home performance a visible and viable industry nationwide.

Multiple Business Models

There are probably as many potential home performance business models as there are contractors. However, there are two end points of this continuum that are useful to examine: the home performance consultant and the home performance contractor. The consultant provides a third-party evaluation, recommendation, and perhaps postinstallation verification, but he or she doesn't install the improvement measures. The contractor does it all. For both contractors and program administrators, there are opportunities and challenges with each approach (see "The Consultant Versus the Contractor"). I believe that the comprehensive contractor approach has a greater advantage in the market right now. However, there are a variety of reasons for choosing one model or the other, or for choosing something somewhere in between. The important thing is to recognize which approach will be most effective in a given area and to design the business or program accordingly. In a pilot kicking off in Minnesota, where a rater infrastructure already exists, the proximity to Wisconsin's consultant-based approach to Home Performance with Energy Star has led program administrators to center their efforts on consultants. In places like Atlanta and St. Louis, for reasons that include local infrastructure and high-quality remodeling contractors who are well positioned to take the step to home performance, start-up programs have focused on developing home performance contractors.

Another opportunity that Wisconsin has developed with very encouraging results is hooking consultants up with remodeling contractors. Originally, several programs looked at remodelers as potential home performance contractors. And some remodelers are making the transition with amazing results. However, Wisconsin has shown that remodelers can also be attractive clients of home performance consultants and contractors. Dozens of remodelers in Wisconsin are using third-party consultants to identify potential problems, make recommendations, and verify the quality of their subcontractors' work. And they are emphasizing the independent quality assurance (QA) that a third-party evaluator provides in their marketing approaches to homeowners.

Contractors Are the Key

Regardless of the business model, it is the contractors' initiative that makes these market transformation efforts fly. Programs can provide training opportunities for contractors and market the heck out of a concept. But this alone is not enough. Contractors who sit back and wait for even robust program marketing efforts to generate leads and keep them busy are unlikely to thrive in today's market. Conversely, contractors who really integrate the home performance approach into their business, and then market their business as they normally would, are more likely to find success. This is not to say that program level marketing isn't very important. It absolutely helps prime the pump and establishes a high level of credibility for home performance contracting. Real go-getter contractors can achieve success without heavy marketing at the program level. But the more programs are able to support contractors' efforts, the faster their businesses will grow.

One recurring theme I've observed during the growth of the different local programs is that it's a lot easier to teach contractors the technical skills than it is to teach them basic business skills and the nuances of marketing, selling, and delivering comprehensive services. This realization may have big implications for how programs structure entry incentives for contractors and how they go about recruiting contractors to participate in the program. For example, managers of a developing program in Colorado are taking a hard look at how best to use their resources to ensure that the contractors they invest time and energy in are likely to run a home performance contracting business successfully. To that end, they're targeting contractors who value the approach enough that they are willing to pay for their own training and equipment, and to commit staff to the necessary training and mentoring. This required investment forces contractors to self-screen for serious commitment and allows the Colorado program to focus their efforts where they are most likely to succeed.

Tailoring the Message

To help consumers embrace the concept, marketing and sales need to be responsive to consumer needs. Programs and contractors alike report that energy or cost savings aren't usually the primary motivators for homeowners. These customers seek out home performance contractors for a variety of reasons, ranging from comfort to building durability to health and safety. This may change as energy prices rise. But it's clear that what's important in designing consumer marketing pieces is listening to homeowners and responding to their needs. The great thing about home performance is that applying good building science allows contractors to solve a variety of problems—including problems that may have been previously misdiagnosed—and save energy in the process.

Crucial Financing

Whether contractors are participating in programs, or whether they're on their own like Joe Kuonen in Arkansas, they all identify financing as critical. Dick Kornbluth, a contractor in Syracuse, New York, says, "Four years ago almost all of my customers paid cash. Now most of them use financing." Why the change? One of the primary reasons is the scope of the jobs. Several years ago, most of Kornbluth's jobs were limited to installing insulation, and most people could scrounge up the $1,000-$3,000 they needed to pay for the work. But the scope of the jobs has grown to an average of over $7,000 in New York, with some jobs pushing up as high as $20,000 for energy-related measures alone. Access to financing becomes key to closing these types of deal. In response, some programs offer lower-than-market-rate financing through interest rate buydowns and other mechanisms. In New York, for example, homeowners can get a loan offered through Energy

THE CONSULTANT VERSUS THE CONTRACTOR

	The Home Performance Consultant		The Home Performance Contractor	
	• Independent evaluation, recommendation, and postimprovement inspection but consultant does not install measures • Typical consultants are home energy raters (or other auditors) expanding their offering in the existing-homes market • Consultant refers homeowner to allied contractors, but the homeowner enters into new contracts with the relevant contractors • Profits from fees charged to homeowner (or other client); the evaluation/audit is the product		• One-stop shopping for the homeowner • Diagnostic, evaluation, shell, and HVAC improvement capacity • In-house; or • Subcontract arrangements • Profits directly from work done in home • Usually uses evaluation/audit as a loss leader	
Consultant/ contractor perspective	**Opportunities** • Entry point for raters and others into expanded existing-homes market • Builds on existing skills without large capital investment (relative to contractor model) • Business is easily scalable from one person up; low overhead so don't need large staff • Clients can range from homeowners to remodelers to other contractors	**Challenges** • Biggest hurdle: The market does not place much value on this so how can homeowners be convinced to pay $500 for it? • In the short term, contractors' business rests on fees; in the long term it rests on the results produced by other contractors • Significant training of contractors required can't ignore this just because the consultant is the focus	**Opportunities** • Loss-leader evaluation removes hurdle • Capitalizes on trust relationship to close deal with homeowner • Profit opportunity regardless of the mix of measures • Complete job scope facilitates financing • In-house capacity minimizes intertrade ball dropping	**Challenges** • Most contractors don't start with complete in-house capacity and need to hire and perhaps train subcontractors • Start-up costs (for example, training and equipment) focused in one company • Bigger jobs mean customers now need access to financing
Program Perspective	**Opportunities** • Ability to tap into previously trained auditors and raters • Built-in third-party QA/QC • Possibility of bringing good building science into remodeling projects without having to train every remodeler! • Use of rater may open door for some existing financing products	**Challenges** • Incentive for consultant to move homeowners to make improvements is indirect without program support • Programs will probably need to subsidize the audit • Additional steps between consultant and contractor mean that some homeowners will drop out • Contractors need to be trained to deliver high-quality improvement but they are one step removed from the program	**Opportunities** • Profit motive drives market to installed measures • Leads to comprehensive jobs and greater energy savings thus far, job scopes appear larger than under consultant approach • Homeowners not lost between audit and work • Program not burdened with creating relationships between consultants and contractors (or among contractors)	**Challenges** • May not tap into previously trained contractors, especially raters • Recruitment focus probably means creating new converts, not simply tapping into the converted • Financing! • Currently no market mechanism for QA programs must build it into their program design

Finance Solutions, and the state buys down the interest rate to 5.99%.

All contractors will admit that better interest rates make selling jobs easier. But when pressed, most say that it is not the lower rates that are critical, it is the ease of access. With 24-hour turnaround and the potential to prequalify customers—or even to offer a ten-minute turnaround on some consumer loan and credit card applications—contractors are able to close the deal while they're in the house and the customer is excited about the possibilities. "If homeowners are left having to hunt down their own financing, many will simply drop out, and I'll lose them as customers," says Kornbluth.

Not all contractors have access to easy financing right now, but programs and contractors themselves are working on the question. Some local and regional lenders are stepping in to provide easy and attractive options. Some contractors are exploring opportunities through HVAC equipment manufacturers and credit cards. For example, some credit card vendors offer companies the opportunity to buy down interest rates to as low as 0% for 6 to 12 months. After that, the balance reverts to the market rate; however, with fast approval and an attractive interest rate to homeowners, this may prove to be a valuable—if expensive—tool. We already see this frequently associated with large purchases in the retail sector and "six-month same as cash" financing. Financing is an area that requires further exploration and broadly applicable solutions.

Standards, Certifications, and Protocols

Currently there are no national consensus standards or nationally recognized certifications for whole-house retrofits. Programs and contractors draw from a wide array of existing standards and expertise to put together their packages of what should be done—and how—in existing homes. And the programs have each had to come up with new ways of making it easy for the consumer to identify qualified contractors. This is beginning to change. Working in concert with New York's Home Performance with Energy Star program, the Building Performance Institute (BPI) has begun to fill this need through its certification and accreditation program (see "Moving Forward with Building Performance," p. 6). The EPA, with funding also provided by HUD and DOE, last fall announced a $1 million grant to BPI to advance the development of a national infrastructure of certified technicians and accredited contractors to deliver whole-house energy efficiency improvements.

Beyond standards, programs and contractors alike could benefit from comprehensive protocols addressing all aspects of whole-house improvements. The California Energy Commission (CEC) has funded the development of comprehensive protocols covering the full range of home diagnostics, corrective actions, and business practices. Although the protocols are still in draft form, they are already starting to affect program design. A parallel home performance contracting implementation program funded by the California Public Utilities Commission is field-testing them, and start-up programs elsewhere are considering them as a way to create a structure without having to reinvent the wheel. The final CEC protocols are expected to be available by the end of the first quarter of 2005.


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Air sealing is a crucial step in creating a high-performance home. Here large bypasses in an attic are sealed off.

Filling the Training Pipeline

The need for good business skills notwithstanding, as contractors grow their businesses and the industry grows, there will also be an increasing need for skilled technicians. A lot of training will probably be done in-house. But will it be enough? Currently, a major bottleneck to increased production is the lack of crews to do the work. Successful contractors report that they can sell a lot more work than they can deliver. One solution is to build the infrastructure and increase the pool of technicians to draw from. In New York, New York State Energy and Research Development Authority (NYSERDA) has worked with a vocational school to develop training curricula that feed into BPI certifications. In Missouri and elsewhere, program managers and even participating contractors are looking at these curricula and other resources to explore the possibility of bringing building science-based training more broadly to community and technical colleges.

We've already learned that home performance contracting works in the market. We know Home Performance with Energy Star can help contractors bring the concept to their local market more quickly. And we've learned a lot of lessons about how programs can tune this message to their local markets. We still have a lot more to learn, but we're on our way to establishing a truly national industry.

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FOR MORE INFORMATION:

To learn more about Home Performance with Energy Star, visit www.energystar.gov/homeperformance.

A NEW YORK ENERGY $MART PROGRAM

Creating a sustainable business model for large-scale deployment of energy efficiency services into the existing-homes sector has been a daunting challenge for several decades. The New York State Energy Research and Development Authority (NYSERDA) responded to this challenge by launching Home Performance with Energy Star, a comprehensive program aimed at improving the energy efficiency, comfort, affordability, and safety of existing one- to four-family homes in New York State. Within the program, qualified contractors diagnose homes and address the identified concerns, such as indoor air quality and comfort, while also using energy efficiency and preventive-maintenance techniques to save the homeowner money. For New Yorkers seeking to improve the overall performance of their homes, the program brings a one-stop shopping experience to home improvement. The program was initially launched in six target markets: Albany, Buffalo, Rochester, Syracuse, Binghamton, and the Hudson Valley. Expansion into the New York City, Long Island, and Westchester markets proceeded through 2004.

This program transforms New York's energy-efficient construction marketplace through a comprehensive whole-house training curriculum that leads to mandatory contractor certification, through BPI, coupled with a comprehensive consumer and contractor awareness/education campaign, and incentives aimed at overcoming barriers for contractors and retailers. The program also offers training around New York State through the Onondaga-Cortland-Madison Board of Continuing Education Services (OCM BOCES), a vocational education provider, to assist contractors in preparing for the BPI certification tests. All training offered through the program is subsidized by NYSERDA.

In addition to building a well-trained, professional home performance contractor infrastructure, NYSERDA strove to drive up consumer demand for these services. To that end, the agency developed an aggressive call-to-action marketing campaign focused on three crucial areas:

1. recruiting and educating contractors to effect change in home improvement services by using a whole-house approach for diagnosing and treating homes;

2. increasing consumer awareness of and demand for the services offered by participating Home Performance with Energy Star contractors; and

3. providing a small consumer incentive in the form of interest rate reduction through either an unsecured Fannie Mae Home Improvement loan or a secured New York Energy $mart Loan.

NYSERDA has also launched the New York Assisted Home Performance with Energy Star program, which provides subsidies to income-eligible New York households, who may not qualify for the Weatherization Assistance program, to complete energy efficiency upgrades to their homes.


NYSERDA

A mass media marketing campaign—using Steve Thomas (pictured above), host of This Old House, as celebrity spokesperson—brought the program immediate credibility and recognition, which helped to generate quick consumer demand.

Initiated in 2001, New York's Home Performance with Energy Star program is already showing promising results:

• Over 5,800 jobs have been completed at an average of $7,000 per job to date. Residential customers have invested more than $40 million of their own money in home energy improvements. NYSERDA has contributed an additional $8 million in subsidies to help more than 2,100 income-eligible households pay for installation of eligible measures under the New York Assisted Home Performance with Energy Star program.

• More than 330 technicians have been certified and over 100 contracting firms have been accredited through BPI in whole-house building diagnostics and proper installation of measures for greater energy efficiency, health, and safety.

• Consumer awareness of Energy Star products and services has increased as a result of NYSERDA's marketing campaign and its cooperative advertising program with contractors, in coordination with its marketing efforts for other Energy Star programs that it sponsors. New York now ranks among states with the highest level of Energy Star awareness in the nation, and understanding of the label in New York has increased from 20% in 1999 to 60% today.

Because this program takes a unique approach—an approach that differs greatly from the approach taken by conventional rebate-driven energy efficiency programs—it has taught stakeholders a number of interesting lessons. Here are three of the most important ones:

Start small. By launching this program in a single market, and then proceeding to expand it one market at a time, program implementers and NYSERDA were able to integrate modifications quickly and effectively.

Market big. In order for this market-based program to succeed, it was crucial to strike a balance between consumer demand and contractor infrastructure. The call to action mass media marketing campaign—using Steve Thomas, host of This Old House, as celebrity spokesperson—brought the program immediate credibility and recognition, which helped to generate quick consumer demand. This aggressive and extensive marketing campaign also served to convince potential participants in the contracting field that NYSERDA was making a long-term commitment to the program.

Provide quality assurance. As the program reached full-scale implementation, it was necessary to provide permanent systems of quality assurance in order to maintain the program's credibility. Supplementing the BPI on-site inspections through our implementation contractor allows us to score all contractors in the program for quality. This not only protects consumers in this relatively new field but also creates a level playing field for the competent and professional contractors engaged in the program.

WISCONSIN'S HOME PERFORMANCE WITH ENERGY STAR

Wisconsin's homeowners can now tap into a network of highly trained building consultants and contractors to improve the energy efficiency, safety, and durability of their most important investment—their homes. Over the past three years, Wisconsin's Home Performance with Energy Star program has constructed this network of building professionals. It started with an existing pool of highly skilled home performance consultants and later added insulation, HVAC, and remodeling contractors to support them. The program's success is illustrated by the phenomenal growth in the number of completed home performance jobs and participating contractors (see Table A).

For each new project, a home performance consultant evaluates the home, writes a detailed report, and recommends the appropriate qualified contractors. In another measure of the program's success, an increasing number of homeowners have been following through on consultants' recommendations. In the past year, 73% of all homes evaluated have had at least one energy efficiency measure installed, as compared with less than 20% in the first year.

Once contractors have completed the work, the consultant returns to the home to perform an inspection and conducts a second round of blower door and combustion safety tests to verify the results. Homeowners are assured of high-quality work, and they appreciate the unbiased third-party verification and performance testing offered by the consultant.

In this process, consultants form relationships with contractors that result in a fully functional network of service providers, all of whom are committed to doing high-quality work. Consultants work for the company, but also for the consumer, since any callbacks resulting from quality issues affect the profitability of the consultant as well as the contractor. Similarly, all of the companies within a network comarket and refer work, so the quality of the work reflects on all of the companies involved.

Projects can originate from anywhere in the network and referrals are shared throughout the network. Experience shows that once a consultant starts to develop his or her network, word-of-mouth referrals follow. And as a consultant completes more home performance jobs, the scope of the work becomes more varied, additional contractors and different trades become involved, and additional referrals are generated.

Table A. Wisconsin's Program by the Numbers

	Number of Jobs	Participating Contractors	Annual Therm Savings*	Annual KWH Savings*
Year 1 ('01–'02)	97	32	52,000	145,000
Year 2 ('02–'03)	422	57	169,000	370,000
Year 3 ('03–'04)	1,063	168	545,000	1,163,000

* Savings are accumulated from deemed savings. Deemed savings are engineering calculations developed with staff and state evaluators.

Currently, the Wisconsin Home Performance with Energy Star program is part of Focus on Energy, Wisconsin's statewide energy efficiency and renewable energy initiative. But home performance activities in Wisconsin began before the 2001 launch of Focus on Energy. An earlier state-sponsored Home Performance Rating program, managed by the Wisconsin Energy Conservation Corporation, identified, developed, and trained highly qualified building science professionals. With this network of professionals in place three years ago, it made sense to build Focus on Energy's Home Performance with Energy Star program around this core group.

With the consultants in place, the new program needed one more element to succeed: qualified contractors to install the recommended improvements. This final step presented big challenges: It required assembling, training, and maintaining a group of reliable contractors. But persistence paid off. The Wisconsin Home Performance with Energy Star program now operates successfully using its two most important tools: dedicated consultants and contractors that meet the needs of Wisconsin's homeowners.